CLASS 10 · CBSE · NCERT MATHEMATICS · PYQ

Chapter 14: Probability

Previous year board questions, topic-wise: 21 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Complementary events

1. A die is rolled once. Find the probability that the number shown is not 3. (a) $\frac{1}{6}$ (b) $\frac{3}{6}$ (c) $\frac{5}{6}$ (d) 1 CBSE 2026, 1 mark

2. A player wins a game with probability 0.79. What is the probability that the player loses it? (a) 1.79 (b) 0.31 (c) 0.21% (d) 0.21 CBSE 2024, 1 mark

3. An event happens with probability p and does not happen with probability q. How are p and q related? (a) $p+q=1$ (b) $p=1, q=1$ (c) $p=q-1$ (d) $p+q+1=0$ CBSE 2023, 1 mark

4. In a group of 20 people, 5 cannot swim. One person is chosen at random. Find the probability that the person can swim. (a) $\frac{3}{4}$ (b) $\frac{1}{3}$ (c) 1 (d) $\frac{1}{4}$ CBSE 2023, 1 mark

5. In a game a coin is tossed 3 times. A player wins if all three tosses give the same result. Find the probability of losing. CBSE 2019, 2 marks

6. An integer is chosen at random from those lying between 1 and 100. Find the probability that it is (i) divisible by 8 (ii) not divisible by 8. CBSE 2018, 2 marks

2 Theoretical probability

7. Assertion (A): The probability that a leap year has 53 Mondays is $\frac{2}{7}$. Reason (R): The probability that a non-leap year has 53 Mondays is $\frac{5}{7}$. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2026, 1 mark

8. Assertion (A): When one number is picked at random from the numbers 1 to 20, the probability that it is a number from 1 to 20 is 1. Reason (R): An event E with $P(E)=1$ is a sure event. Choose: (a) both true and R explains A (b) both true but R does not explain A (c) A true, R false (d) A false, R true CBSE 2025, 1 mark

9. The even numbers are removed from the list 1, 4, 7, 9, 16, 21, 25, and one of the remaining numbers is picked at random. Find the probability that it is prime. (a) $\frac{2}{5}$ (b) $\frac{1}{5}$ (c) $\frac{1}{7}$ (d) $\frac{2}{7}$ CBSE 2024, 1 mark

10. 6000 lottery tickets are sold. A girl works out that her probability of winning the first prize is 0.08. How many tickets did she buy? (a) 40 (b) 240 (c) 480 (d) 750 CBSE 2023, 1 mark

11. A bag holds 4 red, 3 blue and 2 yellow balls. One ball is drawn at random. Find the probability that it is (i) red (ii) yellow. CBSE 2023, 2 marks

12. A die is thrown once. Find the probability of getting (i) a prime number (ii) a number lying between 2 and 6. CBSE 2019, 2 marks

13. Three different coins are tossed together. Find the probability of getting (i) exactly two heads (ii) at least two heads (iii) at least two tails. CBSE 2016, 3 marks

14. A number x is chosen at random from 1, 2, 3, 4, and a number y is chosen at random from 1, 4, 9, 16. Find the probability that the product xy is less than 16. CBSE 2016, 4 marks

3 Two dice

15. Two dice of different colours are thrown together. List all the possible outcomes. Find the probability that (i) both dice show the same number (ii) the dice show different numbers. CBSE 2026, 3 marks

16. Two dice are thrown together. Find the probability that the two numbers differ by 2. CBSE 2025, 3 marks

17. Two dice are thrown together. Find the probability that the two numbers add up to 2, 3 or 5. (a) $\frac{7}{36}$ (b) $\frac{11}{36}$ (c) $\frac{5}{36}$ (d) $\frac{4}{9}$ CBSE 2024, 1 mark

18. Two different dice are thrown together. Find the probability of getting (i) a doublet (ii) a sum of 10. CBSE 2018, 2 marks

4 Playing cards

19. One card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is a red face card. CBSE 2025, 1 mark

20. A black card is missing from a pack of 52 playing cards. One card is drawn at random from the rest. Find the probability that it is the queen of hearts. CBSE 2024, 2 marks

21. A card is drawn at random from a well-shuffled pack of 52 cards. Find the probability that it is neither a red card nor a queen. CBSE 2016, 1 mark

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